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Evaluating Quality of Visual Explanations of Deep Learning Models for Vision Tasks

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***Abstract*—Explainable artificial intelligence (XAI) has gained considerable attention in recent years as it aims to help humans better understand machine learning decisions, making complex black-box systems more trustworthy. Visual explanation algorithms have been designed to generate heatmaps highlighting image regions that a deep neural network focuses on to make decisions. While convolutional neural network (CNN) models typically follow similar processing operations for feature encoding, the emergence of vision transformer (ViT) has introduced a new approach to machine vision decision-making. Therefore, an important question is which architecture provides more human-understandable explanations. This paper examines the explainability of deep architectures, including CNN and ViT models under different vision tasks. To this end, we first performed a subjective experiment asking humans to highlight the key visual features in images that helped them to make decisions in two different vision tasks. Next, using the human-annotated images, ground-truth heatmaps were generated that were compared against heatmaps generated by explanation methods for the deep architectures. Moreover, perturbation tests were performed for objective evaluation of the deep models' explanation heatmaps. According to the results, the explanations generated from ViT are deemed more trustworthy than those produced by other CNNs, and as the features of the input image are more dispersed, the advantage of the model becomes more evident.**

***Index Terms*—Explainable artificial intelligence, Vision Transformer, heatmaps, subjective evaluation.**

I. Introduction

Explainable artificial intelligence (XAI) [1] can determine whether computer vision (CV) models are effective and can learn meaningful concepts from the data. Using XAI, the outputs of deep learning models can provide specific reasoning along with their decisions instead of merely outputting predictions with probability scores, thereby allowing users to confirm that the decision made by the model is based on plausible features rather than spurious correlations or artifacts [1].

Human-understandable explanation plays a crucial role in the quality assessment (QA) of imagery for machine integration purposes where human-machine interaction is needed for decision making. In this case, high-quality explanations of the machine decisions must be provided in a manner that is interpretable by humans. Unlike the conventional QA goals that aim to maintain the visual appeal of the image, here, the focus is on delivering accurate, explainable, and reliable information to enhance the interaction.

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Several methods have been proposed to use visual heatmaps to highlight the relevant features of input images, aiming to explain the decisions of deep learning models. Gradient-based methods [2]–[5] and attribution propagation methods [6]–[8] are the most widely explored XAI categories. Alternative XAI methods include saliency-based methods [9], perturbation methods [10], and Shapley-value methods [11]. Meanwhile, with various XAI methods being proposed, evaluating the quality of explanations has become an active field of research. Current evaluation methods are primarily categorized into objective evaluations and human-centered evaluations. Objective metrics [12], [13] are mainly used to assess the stability and robustness of explanations. Human-centered evaluations [14] investigate the effectiveness of XAI methods, i.e., whether heatmaps can build a complete and correct inferential process of a given model that matches the human mental and semantic representation. However, existing work [13], [14] pays attention to the comparison of different XAI methods while ignoring the differences among deep learning models that are strongly tied to the explanation methods.

With the introduction of the vision transformer (ViT) [15], the landscape of network architecture design for CV tasks has been altered dramatically. Under certain requirements, such as larger dataset sizes and higher computational cost, ViT (and the recent Swin Transformer [16]) can outperform standard convolutional neural network (CNN) models, e.g., ResNet [17], on machine vision tasks. Due to their impressive performance, transformers can displace CNN in CV. Yet, due to the complexity of transformers, CNNs remain attractive. ConvNeXt [18], which is inspired by the design of transformer architecture but constructed entirely from standard CNN modules, is shown to compete with transformers. ConvNeXt not only maintains a fully-convolutional operation, making it simple to implement but also has an excellent performance.

Considering the variety of deep architectures, it is necessary to examine the quality of explainability, that is, which models' decisions are most in line with human inferences. It is likely that CV models that can closely mimic the human vision can make decisions that are more understandable to humans. The most effective method of evaluating whether a model can provide human-friendly explanations is through subjective tests. Such studies [5], [19] typically ask participants which explanation they found more reasonable. Recently, related human studies have been conducted [14], [20]. However, these

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methods present users with explanations and do not take into account that users may learn from these explanations, potentially leading to bias in their judgment. Different from them, this paper aims to create a ground-truth visualization based on the user's visual perception. To this end, we design a subjective experiment that collects human annotations and then assess the correlation of the ground truth with AI explanations. The closest to our work is [21], but they require eye-tracking equipment and have conditional restrictions on the eye-trackers; hence, our method is easier to implement.

The contribution of this paper is as follows: (i) we developed a simple and efficient subjective experiment to collect human annotations as benchmarks. (ii) we deployed metrics to calculate the overlapping rate of the important features of explanations, which plays an important role in determining whether the distribution of key information in the explanations is located in the same place. (iii) Through two CV tasks, we conducted a comparative evaluation of the explainability of CNN-based and transformer-based models in terms of both objective and subjective evaluations. The results indicate that ViT outperforms CNN-based models in terms of explaining dispersed local features on the image.

II. Background on Visual Explanations

A. Visual Explanations for CNNs

XAI methods for CNNs highlight important features in the input image that contribute most to a particular classification decision made by the model. Specifically, Gradient⋆Input [2] is the first gradient-based XAI method that has been proposed. The basic idea is to compute the attribution by taking the partial derivatives of the output with respect to each input and multiplying them feature-wise by the input itself. Integrated Gradients [3] is proposed as a generalization in which the average gradient was used for the nonlinearities in the model. SmoothGrad [4] adds random Gaussian noise to the input image at each iteration improving the smoothing performance of the visualization. Gradient-weighted Class Activation Mapping (Grad-CAM) [5] is a widely used class-specific approach, which uses the gradients of any target concept flowing into the final layer to produce a coarse heatmap for visualizing the regions in an input that contributes most to a particular classification decision made by the network architecture.

Additionally, propagation-based explanations explain the model prediction by progressively redistributing the output on the lower layers via propagation rules. Examples include Layer-wise Relevance Propagation (LRP) [6], which propagates relevance backward from the decision class until the input variables have been reached based on the Deep Taylor Decomposition (DTD) principle [7], DeepLIFT [8], which is similar to LRP.

B. Visual Explanations for Transformers

The current landscape of visualization methods for transformers is not as extensive as that of CNNs. Attention visualization depicts the attention weights that the model assigns to each token in the input sequence when making a prediction. This includes self-attention, which uses the attention map of the first block to extract the relevance, and the roll-out method [22], which includes all the attention maps computed along the forward pass. LRP [6] is also applied to transformers: it considers the information flow through the entire network but has a limitation in that no relevant scores are propagated back to the input. Grad-CAM [5] is also applicable to transformers. As mentioned in Section II-A, the computation of Grad-CAM is related to the last convolutional layer in a CNN model; hence, performing Grad-CAM in transformers should set all tokens in the output tensor as zeros except the classification token. In [23], the authors employed LRP-based relevance to compute scores for each attention head in layers of a transformer which obtains state-of-the-art results, although, this method is not applicable to CNNs.

Among contemporary XAI approaches applicable on both architectures, Grad-CAM can be deployed as a common explanation method to investigate which model can generate more trustworthy and human-friendly visual explanations.

III. Methodology

This section describes our subjective evaluation method for assessing whether visual explanations of CNNs and transformers correlate with the HVS.

A. Visual Explanations via Grad-CAM

We start by briefly describing Grad-CAM, which is used as the XAI method for our analysis. The Grad-CAM heatmap $\boldsymbol{L}^c_{\mathrm{Grad-CAM}} \in \mathbb{R}^{w \times h}$ for CNNs is generated by finding the relationship between the target class and the activations in the final convolutional layer. Both the image of width w and height h and the target class c are required as inputs:

$$\boldsymbol{L}^c_{\mathrm{Grad-CAM}} = \mathrm{ReLU}\Big(\sum_{i,j} \frac{\partial y^c}{\partial A^k_{ij}} \cdot A^k_{ij}\Big), \tag{1}$$

where A^k_{ij} is the activation at location (i, j) of the feature map $A^k \in \mathbb{R}^{w \times h}$. The backpropagating gradients of the score for the class c with respect to these activations can be represented as $\frac{\partial y^c}{\partial A^k_{ij}}$, and they are global-average-pooled over the input image dimensions to obtain the neuron importance weights. Grad-CAM is then obtained as a linear combination of weights and activation maps followed by a ReLU function.

Similar to its use in CNNs, Grad-CAM is applied to the last attention layer of the transformer to generate an explanation. But it is important to note that the feature map obtained from the transfomer have a dimension of $A^k_T \in \mathbb{R}^{m \times n \times n}$, where m is the number of heads and n is the number of the input tokens instead of a 2D feature map. Thus, to obtain the final visualization, the output from transformer should be reshape to $\sqrt{n-1} \times \sqrt{n-1}$.

B. Subjective Evaluation Method

We now describe the design of our subjective evaluation method. When doing a classification task on a given image dataset, there exists a set of distinctive features in each image that distinguish its class. These features can be defined as

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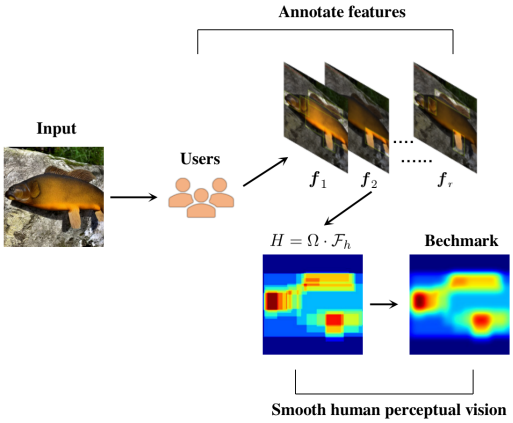


Fig. 1: The process of creating a subjective annotated heatmap.

$F = [f_1, f_2, \ldots, f_n]$. A subset $F_m \subseteq F$ of features can be extracted by a neural network. Only a subset of these features, $\mathcal{F}_m \subseteq F_m$, are highlighted by the XAI methods when forming heatmaps that explain the network outputs. Moreover, a subset $F_h \subseteq F$ of features are perceived by humans when determining the class of the image. We hypothesize that machine-generated human-understandable explanations are heatmaps that overlap with the human visual perceptual features, i.e., $F_h \cap \mathcal{F}_m \neq \emptyset$. Moreover, the more features overlap the more trustworthy machine-generated explanations become. Conversely, in case $F_h \cap \mathcal{F}_m = \emptyset$, features captured by machine-generated explanations are not perceptible by humans, alias, the explanations are not human-understandable.

To evaluate the visual explanations from different networks, the first step is to collect a human visual feature set $\mathcal{F}_h \subseteq F_h$ in the given dataset D, which serves as benchmark. Fig. 1 illustrates the process of creating such a subjective annotated heatmap. Users are shown an input image $I \in D$ together with its class label. Then, they are asked to annotate class-related features (i.e. those image regions that better help them to make a classification decision) using several bounding boxes. Feature sets for the input image from multiple people are collected, which are denoted by $\boldsymbol{f}_r = [f_1, f_2, \ldots, f_\ell]$, which $\ell \leq n$, and r is an integer presenting the number of human subjects. Combining these feature sets together, a rich and comprehensive set of features $\mathcal{F}_h$ is obtained:

$$\mathcal{F}_h = \boldsymbol{f}_1 \cup \boldsymbol{f}_2 \ldots \cup \boldsymbol{f}_r. \tag{2}$$

As the number of users increases and more subsets $\boldsymbol{f}$ are created, the closer the features contained in $\mathcal{F}_h$ are to those in F_h. Meanwhile, the number of appearances of each feature in the subsets is counted as t and used to calculate the weight $\omega = t/r$. Based on the features $\mathcal{F}_h$ that users have annotated and the weights $\Omega = [\omega_1, \omega_2, \ldots, \omega_\ell]$, the heatmap H indicating the human perceptual vision can be drawn:

$$H = \Omega \cdot \mathcal{F}_h, \tag{3}$$

where the features with high weight values are highlighted in warmer colors. Moreover, to achieve the heatmap display that is nearer to human eyes' observation, Gaussian Blur is applied to H to smooth the edges of the rectangle. Finally, we get the human-understandable benchmark for I. For all images in D, a set $\boldsymbol{H} = [H_1, H_2, \ldots, H_s]$ can also be obtained.

Model	IN-1k acc.(%)	DeepFakes acc.(%)
ViT-B	81.07	99.72
ConvNeXt-B (*iso.*)	**84.06**	**99.96**
ResNet50	76.13	89.25
VGG16	71.59	94.43

TABLE I: The prediction accuracy results on the validation set for ViT-B, ConvNeXt-B, ResNet50, and VGG16 on object detection task and Deepfake detection task.

IV. Experimental Settings

We experiment with the object detection task and Deepfake detection task considering different deep models. For the object detection task, the ImageNet-1K dataset is used, containing 1000 object classes with 1.2M training images. For the Deepfake detection task, we use a subset of Flicker-Faces-HQ dataset, which contains 30K images as real ones and the same number of fake images generated using StyleGAN [24]. This dataset is named DeepFakes. In both datasets the images are resized to 224×224.

A. Model Training and Explanations

In this section, we describe the implementation details for training the models on different tasks. Four models are selected in our experiments, that is, three CNN models (isotropic ConvNeXt-B [18], ResNet50 [17], and VGG16 [25]) and the ViT-B [15] model. The isotropic ConvNeXt-B model is set to use the same feature dimensions as ViT-B. In our experiments, the input of these two models is a sequence of all non-overlapping patches of the input image with size 16×16.

For the object detection task, we directly use the ConvNeXt-B and ViT-B models, VGG16 and ResNet50, at resolution 224×224 pixels from the Pytorch pre-trained models library. The top-1 accuracy on the validation set of each model is reported in Table I. For the Deepfake detection task, we train four binary classifiers with 24K real images and 24K fake images as training set. Due to its limited size, the per-trained ConvNeXt-B and ViT-B models were fine-tuned on the DeepFakes training set (not trained completely). Similarly, fine-tuning was used to train the classifiers for VGG16 and ResNet50. All models are implemented using Pytorch and trained using the ADAM optimizer with a batch size of 64. The learning rate is set to $2e-5$ for ConvNeXt-B and ViT-B, and $1e-3$ for VGG16 and ResNet. The prediction accuracy of all models is reported in Table I.

After the models are trained, we input an image, the corresponding class, and the target layer from the model into the PyTorch grad-cam package [26], to obtain the heatmap generated by Grad-CAM algorithm.

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B. Evaluation Setting

We perform objective and subjective evaluations, with the aim of comparing the stability, robustness and human understanding of the generated explanations.

Objective evaluations. The positive perturbation test is the main objective method to measure the quality of explanations. We input the images in the validation sets of ImageNet-1K and DeepFakes to the different trained models and obtain explanations using Grad-CAM. We then use the heatmaps as guide to gradually remove important pixels from the images, input again the processed images into the trained models and compute the variation in the prediction accuracy. We use the area-under-the-curve (AUC) metric to determine which model highlights the important features that influence the classification score. The pixels are removed from the highest to the lowest relevance and the AUC score becomes smaller. The scores are reported on Table II.

Subjective evaluation. We use the Roboflow platform [27] for collecting human-annotations. The platform provides an annotation tool that records the relative locations of the bounding boxes to the input image. We select 20 images with obvious features from ImageNet-1k and DeepFakes respectively as the inputs. For the ImageNet-1k dataset, obvious features refer to unique parts or shapes of objects associated with a class in an input image that can be easily detected and help humans identify the category of the object. For the DeepFakes dataset, obvious features are artifacts on the image that help humans recognize the input as fake. We invite 10 users (7 men and 3 women) to annotate 40 selected images for the two tasks, that is, $r = 10$ in (2). More than half of these participants have prior experience with CV. They were all highly educated (college graduates or higher education). In total, 400 annotated images are received, i.e. 400 sets of features $\boldsymbol{f}$, of which 200 sets are used for the detection task and 200 sets are used for the Deepfake detection task. Then, based on the method described in Section III-B, some sample human annotations are shown in the second columns of Fig. 2(a) and Fig. 2(b).

In addition, we use distribution-based metrics to analyze which model offers the best approximation to human observations. The human annotations are denoted as H.

- The similarity metric [28], SIM, measures the similarity between two heatmaps:

$$\begin{aligned} SIM(P,H) &= \sum_i min(P_i, H_i) \\ where \sum_i P_i &= \sum_i H_i = 1 \end{aligned} \tag{4}$$

  where P is a given explanation from a model and i indexes the pixel location. The metric is sensitive to the choice of the Gaussian sigma parameter, which we used to smooth human annotations.
- The Person's Correlation coefficient metric [28], CC, is used to measure the linear relationship between heatmaps:

$$CC(P,H) = \frac{\sigma(P,H)}{\sigma(P) \times \sigma(H)} \tag{5}$$

  where $\sigma(\cdot)$ is the covariance function. Locations with high positive CC values indicate that they have values of similar magnitudes.
- The overlapping metric (OL) examines whether the important features highlighted by the heatmaps are consistent. A threshold θ is required as input for this metric:

$$OL(P,H) = \frac{\sum_i (P_i > \theta \wedge H_i > \theta)}{w \times h} \tag{6}$$

  where w and h are the width and height of the input respectively. The threshold θ ranges from 0 to 1 for normalized heatmaps. For our experiments, $\theta = 0.5$.

V. Results and Discussion

A. Analysis of the Results

Table I presents the prediction accuracy results on the validation sets of ImageNet-1K and DeepFakes for ViT-B, isotropic ConvNeXt-B, ResNet50, and VGG16. The results demonstrate that ConvNeXt performs best on both tasks, whereas, ViT is almost on par with ConvNeXt in the Deepfake detection task. Therefore, ConvNeXt remains competitive with ViT in terms of model prediction accuracy.

In addition to accurately predicting results, the explainability of the model to help humans understand the prediction process, is also an important factor in industrial applications. To provide visual explanations for the different models, Grad-CAM is used vis-à-vis the human-annotated benchmark. Sample results are shown in Fig. 2(a) for the object detection task and Fig. 2(b) for the Deepfake detection task.

First, we discuss the quality of the XAI heatmaps from a qualitative perspective. As we can see, in both tasks, the explanations from the ViT model deliver detailed information, in other words, the highlighted parts in the heatmaps are small but distributed. On the contrary, the highlighted regions of the CNN model-based explanations are concentrated. ResNet50-based explanation, in particular, cover a large area that highlights the object as a whole in the object detection task. Fig. 2(a) also displays the scores of the explanations compared to human annotations based on similarity (SIM), correlation (CC) and overlap rate (OL) of importance features metrics. It appears that the explanations generated from ResNet50 are comparable with ViT and in some cases seem to be a bit more aligned with human visual perception, as evidenced by some higher scores obtained in the metrics. However, even if ResNet50 allows more important features to be covered and a smaller AUC score (reported in Table II) to be obtained in the positive perturbation test of the object detection task, this does not necessarily adhere to the HVS. [29] suggests that humans are more sensitive to low-frequency features than to high-frequency features in an image and that we tend to recognize objects by observing and interpreting a sparse set of features points in the field of view in our brain, which is validated (instead of verified ?) by the benchmarks we collected. However, because the CNN models have a propensity to focus on high spatial frequency information [30] and produce large-area explanations that include features not

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	ViT-B	ConvNeXt-B	ResNet50	VGG16
IN-1k	24.75	34.06	19.15	**14.65**
DeepFakes	**8.65**	34.20	29.32	18.74

TABLE II: Positive perturbation AUC(%) results for the target classes on the ImageNet-1K and DeepFakes validation sets. And for positive perturbation lower is better.

perceivable to humans, the explanations derived from them may be unconvincing.

Accordingly, further exploration is required to reach conclusive insights. DeepFakes, a different dataset from ImageNet-1K, is selected to fine-tune the models and serve as inputs for generating explanations. The primary contrast between the two datasets is caused by the distinct dispersion of features associated with specific classes. Specifically, images from ImageNet-1K are classified according to the features of the object, so the features are mainly extracted from the object and are relatively compact, and not as scattered as the artificial traces in the fake images in DeepFakes. From Table II, we see that the explanation for the fine-tuned ViT-B model applied to the Deepfake detection task achieves the best performance by a large margin. Besides, the explanations for the ViT model outperform all three measured metrics.

Thus, we can conclude that as the features of the input image are more scattered, the advantage of the visual explanations from the ViT model becomes more evident, thereby achieving more human-trustworthy AI explanations. And more exploration of the two models for visualization will be discussed in Section V-B.

B. Discussion

Human visual cortex processes information in a hierarchical manner, with sensitivity to complex stimulus characteristics increasing along the ventral visual pathway. ViTs and CNNs also exhibit such multi-scale data processing mechanisms. Recent studies have shown that the encoding of features in deep neural network layers correlates with the processing of data in the visual cortex from the primary visual area (V1) to higher visual areas [31]. Given the correspondence between deep models and the HVS data processing, it would be interesting to discuss the mechanism of the CNNs and ViT with a closer look at their connection to the HVS data processing.

CNNs resemble feed-forward multi-scale feature encoding, but since convolutions are intrinsically local, even deeper layers of CNNs with larger receptive fields, still have difficulty capturing information about global structural relationships [32]. Unlike CNNs, ViTs with their patch-wise self-attention mechanism can provide dynamic receptive fields that span the whole input space [33], which help to achieve better explainability closer to human decisions. Particularly for deep fake detection, where a number of scattered spatial features need to be integrated for a correct determination, ViT provides a much better explanation quality by capturing a broader range of relationships between distorted regions located at different spatial locations. However, CNNs yield less human-like explainability due to the feed-forward-only convolutional mechanism.

ViT's self-attention mechanism can perhaps be compared with—in terms of its ability to provide a more versatile feature encoding—some capabilities of the cortical feedback mechanism in HVS, which plays an integral role in perceptual representations of the HVS, conveying arousal, attention, and context information [34]. According to neuroscience studies, sensory cortical neurons receive synaptic inputs that are not only feed-forward in nature, and the feedback from higher areas can have a significant contribution in the vision tasks. By using feedback projections, neurons may be able to estimate information lacking in the feedforward receptive field and to signal the stimulus features changes across visual space [35]. ViT with its self-attention mechanism, can also go beyond a strict feed-forward mechanism and yields a dynamic receptive field that can grasp the global representation, missing in the feed-forward CNNs.

VI. Conclusion

We design a subjective evaluation method for trustworthy and human-friendly visual AI explanations. Our method devises a benchmark annotated human visual perception and an evaluation metric for calculating the overlap rate between explanations and annotations. The method is used to assess the Grad-CAM explanations of CNN and transformer architectures. We find that the explanations generated from the ViT are more consistent with human annotations, particularly when the feature information is dispersed throughout the image. This is because, unlike CNNs, transformers can create a dynamic receptive field grasping the global representation.

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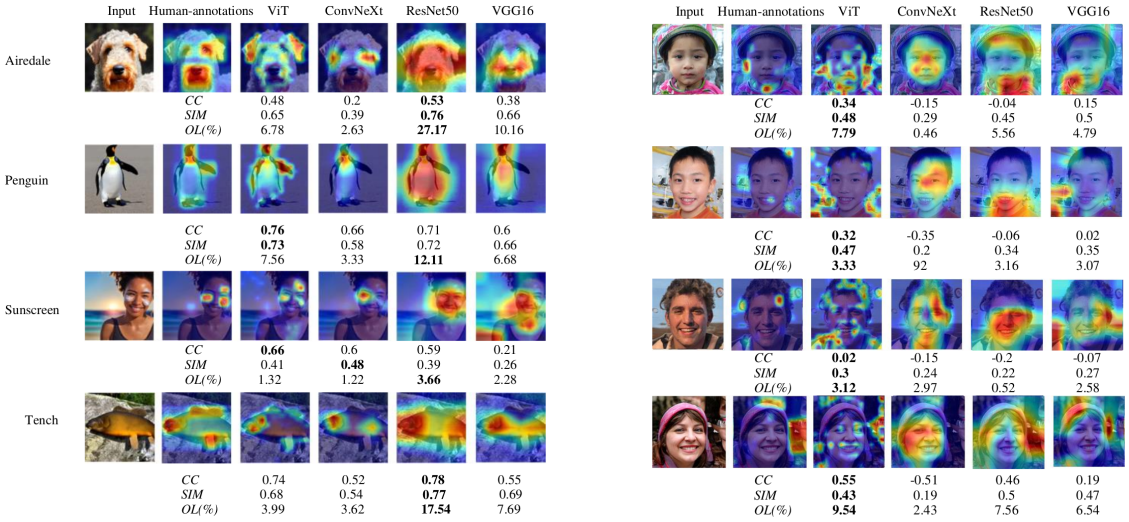


Fig. 2: Sample visual explanations generated by humans and various models for the object detection task (left six columns) and the Deepfake detection task (right six columns and images all are fake). Below each image are three metric scores, which indicate how they compare with human annotations. As we can see, the explanations from ViT perform best on the Deepfake detection task.

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